AROMA PACIFICO

COST DOCUMENT

Cost of the pilot

Coffee traceability pilot for Cauca, Huila and Tolima

Prices of November 2026

This document gathers in one place what the pilot costs the Federation at current prices. It covers the cloud platform in a normal month and in a harvest month, the links of the three collection centers, the development and test environments, inference with its cost per 1,000 invocations in harvest, the hardware and monthly amount of a farm, the drone service and the 100 hours of infrastructure code. It closes with the decisions that lower or avoid cost, the savings the design leaves on purpose, and the assumptions that still need a confirmation. Each section summarizes a detailed cost document of this repository, which holds the calculation line by line.

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1. Terms

Term	Meaning in this document
Normal month	One of the eight months a year with crop monitoring, little drying and a quarter of the sack reception of a harvest month
Harvest month	One of the four months a year with drying at its widest, sack reception at its peak and the harvest estimation flights
Following year	Any year after the first, once the farms are installed and every lot is delimited
Field operation	What the farms cost each month once installed, which is data, SIM charges, maintenance visits, the shared gateways and a reserve to replace parts at the end of their life
Known part	The sum of the items with a verified or estimated price. Items still without a price are listed apart and are never counted as zero
TRM	Tasa Representativa del Mercado, the official rate of pesos per dollar
IVA	Colombian value added tax

2. Summary

Figures in USD before IVA, at the prices of the price basis.

Item	Normal month	Harvest month	Following year
Cloud platform, production	181.78	207.07	2,282.52
Links of the three collection centers	229 to 321	229 to 321	2,748 to 3,852
Development and test, with their keys	12.24	12.24	146.88
Field operation of 200 farms	2,176.88	2,176.88	26,122.56
Drone service	3,777.46	20,410.90	111,864.52
Total without the links	**6,148.36**	**22,807.09**	**140,416.48**
Total with the links	**6,377 to 6,469**	**23,036 to 23,128**	**143,164 to 144,268**

A following year holds eight normal months and four harvest months. The harvest month of the drone service is the one of the first year, and in later years the storage of earlier images adds about USD 0.31 to it. The year column takes the drone service from its own yearly figure, so it is about USD 1 above eight normal months plus four harvest months.

One-time item	USD	COP at the TRM
Field hardware and installation of 200 farms, known part	93,495.54	305,719,196

One-time item	USD	COP at the TRM
Start-up round of the drone service, extra cost of the first year	32,954.95	107,758,732
Infrastructure code, 100 hours	2,276.00	7,442,247

Share of the following year	Percent
Drone service	77.8
Field operation	18.2
Links of the centers	2.3
Cloud platform and environments	1.7

The drone service carries almost four fifths of a year, and the cloud platform less than two percent. Any review of the budget should therefore start with the drone contract and the visits to the farms, and the cloud figures matter mostly as a check that nothing runs without need. The shares use the middle of the link range.

3. Basis of the figures

Choice	Value
Currency	USD, since the cloud bills in dollars and most hardware is priced in dollars
Cloud prices	On-demand list prices of `us-east-1`, with no credit, free tier or discount, read from the AWS Price List API on 10 November 2026
Exchange rate	TRM of 10 November 2026, 3,269.88 pesos per dollar, for every price quoted in pesos
Tax	Every figure before IVA. Section 13 shows the effect of IVA of 19 percent
Month	730 hours for hourly items and 30 days for volumes
Volumes	The pilot figures, 200 farms in Cauca, Huila and Tolima, 60 carriers, up to 400 sacks a day at each of three centers in harvest, and a harvest of four months

The Federation decides which exchange rate and which tax treatment the budget uses. Until it does, the figures follow the price basis, and section 13 shows how much they move.

4. Cloud platform

The monthly cloud cost gives each line with its calculation.

Group of lines	Normal month	Harvest month
Variable lines, which follow the volume of the month	8.08	33.37
VPN and interface endpoints	153.30	153.30

Group of lines	Normal month	Harvest month
Keys, secrets, web firewall, alarms and dashboard	20.40	20.40
Cloud platform, production	**181.78**	**207.07**
Harvest month with all estimation flights		236.76

The variable lines cover telemetry ingestion, carrier positions, sack events, public sack lookups, inference, new drone objects, alert delivery with SMS at its upper bound, the small tables, audit and operational telemetry. Together they stay under USD 34 even in harvest, because every workload runs on services that bill by use and cost almost nothing while idle.

The VPN to the three centers and the private endpoints make up about 84 percent of a normal month. They are what keeps the center traffic off the internet, so the design pays for them. Section 12 shows what removing the endpoints would save and why it is not done.

The last row is the month in which every farm flies its harvest estimation at once. Inference then costs USD 45.95 instead of 16.26, and the platform reaches USD 236.76. The production budget of USD 230 warns at that point, and that warning is expected in such a month.

5. Links of the collection centers

Each center contracts a dedicated link of 80 to 120 Mbps. A Colombian operator publishes reference prices of 250,000 to 350,000 pesos a month for each link, which at the TRM gives USD 229 to 321 a month for the three centers. The reference is indicative and does not say whether it includes IVA, so a quote from an operator with coverage in each municipality replaces it. Shared business fiber, the fallback, costs 70,000 to 150,000 pesos a month for each center, USD 64 to 138 for the three.

These links belong to the centers and are billed by the operator, so they stay outside the cloud budgets.

6. Development and test environments

The development and test environments exist only while the team works on them. At 80 hours a month each they cost USD 9.24, and the keys that protect their state cost USD 3.00 a month whether they are up or not. The environments and promotion design describes how they come up and down.

7. Inference in harvest

Each inference function starts when a complete input arrives, a published flight record for the drone functions and a batch of readings for telemetry. The inference cost counts every billed piece of a run, the person filter, the model, the function time, the queues and, for the harvest estimation, the orthomosaic.

Trigger in a harvest month	Invocations	Images	USD per 1,000 invocations	USD in the month
Focus confirmation	60	3,000	88.72	5.32

Trigger in a harvest month	Invocations	Images	USD per 1,000 invocations	USD in the month
Verification	25	500	35.49	0.89
Harvest estimation	50	5,000	197.93	9.90
Telemetry batch	5,760	0	0.03	0.16
Total	**5,895**	**8,500**	**2.76**	**16.26**

A single cost per 1,000 invocations hides the difference between a telemetry batch and a drone flight, so the table shows it by trigger. Per 1,000 images analyzed, the drone functions cost USD 1.89 in harvest. A normal month runs no estimation flight and costs USD 1.63. Inference runs on the United States inference profile, so crop images leave the country for AWS Regions in the United States, and an image with a person never reaches the model.

8. Farm hardware and monthly amount

The farm hardware and monthly amount prices each part of a node and each visit, in a low, a base and a high scenario. The figures here are the base scenario, with a quarter of the farms on LoRa sharing one gateway for every ten farms.

Item	Fleet of 200 farms	Each farm
Hardware and installation, known part, once	93,495.54	467.48
Data, SIM, maintenance and gateways, monthly	856.67	4.28
Replacement reserve, monthly	1,320.21	6.60
Field operation, monthly	**2,176.88**	**10.88**
Monthly amount proposed for each farm, with its share of the fixed cloud lines		11.77

The investment is the known part. The enclosure, the controller board, the climate and grain probes, the mast, the cables, shipping, import duties and IVA still have no price, and each one raises the investment and the reserve. The monthly amount is a proposal of the team that recovers every recurring cost with no margin, while the investment is presented apart. Whether the farmer pays any of it, and how much the Federation subsidizes, is a decision of the Federation.

The known investment of a cellular farm, USD 451.29, equals about 92 kilograms of dry parchment coffee at the reference price of the Federation of 10 November 2026.

9. Drone service

A contracted operator runs three brigades, one for each department, each with a pilot, a support person, a spraying drone and a mapping drone. The drone campaign cost prices the five services by mission hour from public Colombian labor and equipment rates.

Period	USD	Each farm	COP at the TRM
Start-up month, first year	21,622.25	108.11	70,702,163
Harvest month	20,410.90	102.05	66,741,194
Outside-harvest month, first year	3,321.90	16.61	10,862,214
Outside-harvest month, following years	3,777.46	18.89	12,351,841
First year	144,819.47	724.10	473,542,289
Following years	111,864.52	559.32	365,783,557

A harvest month flies about 140 missions, 40 confirmations, 25 fumigations, 25 verifications and 50 harvest estimations. The first year adds the delimitation of all 200 farms, spread over two start-up months. The agrochemical product stays out, since its dose and area depend on an agronomic prescription, and so does inference, which section 7 already counts.

The operator rates, insurance, permits and logistics are team estimates shown in three scenarios. The low scenario gives USD 75,601 for a following year and the high scenario USD 188,830, so a quote moves this line more than any other.

10. Infrastructure code

The infrastructure code of the network, the security, the ingestion pattern, the inference pattern, the farm module and the pipelines took 100 hours over two weeks. It is a one-time cost.

Profile	Hours	USD an hour	USD
Senior	34	32.00	1,088.00
Junior	66	18.00	1,188.00
Total	**100**		**2,276.00**

The hourly rates are team estimates of the Colombian market. The code is reused for every new center and for every new farm through variables, so growing the pilot adds no new code.

11. Decisions that lower or avoid cost

Each decision is already part of the design, and its effect comes from the detailed cost documents.

Decision	Alternative avoided	Effect, USD
Drone missions only where a sensor alert or a scheduled service calls them, paid by mission hour	Weekly sweeps of every farm	About 2.04 million a year
Drone operator paid by mission hour	Three brigades dedicated full time	88,872.99 a year

Decision	Alternative avoided	Effect, USD
Development and test up only while the team works	Both environments on all month	75.07 a month
Six custom metrics, one dashboard and saved queries	Fifteen metric series and two dashboards	5.58 a normal month, 5.42 a harvest month
Push first, SMS only for urgent drying alerts	SMS for every crop alert too	5.09 a harvest month
Photos and flight logs in Glacier Instant Retrieval after 90 days	Five years in S3 Standard	13.68 a month in year five

Drones directed by the sensors. A drone flies to a farm when a sensor reading opens a case, when a fumigation needs its verification, or once a harvest for the estimation. Sweeping every farm every week with the same rates would cost about USD 2.15 million a year, about 19 times the service. Paying the operator by mission hour, and letting its staff serve other clients between campaigns, avoids the cost of six people dedicated all year, since outside harvest a brigade flies about 0.4 missions a field day. The risk is an operator that asks for a retainer to keep brigades available, so the tender asks for both prices.

Environments on demand. Development and test come up through the pipeline for a work session and come down when it ends. Left on all month they would cost USD 84.31 instead of 9.24. The account budget warns when an environment is forgotten.

Minimal telemetry. The platform keeps as metrics only the six signals an alarm needs and reads the rest with saved queries, which cost almost nothing per scan. A full catalog of fifteen series and two dashboards would cost USD 10.71 in a normal month against 5.13 with the chosen set and its one operations dashboard.

Push before SMS. A crop alert opens a drone case that takes days to resolve, so it travels by push only. An SMS goes out only for an urgent drying alert that the phone did not confirm. Sending an SMS for each of the about 100 crop alerts of a harvest month would add USD 5.09 to that month.

Cold storage for photos. Drone photos and flight logs must stay five years and are rarely read after a case closes. After 90 days they move to Glacier Instant Retrieval, which still returns an object in milliseconds for a claim. By year five the archive holds about 720 GB, and the colder class saves USD 13.68 a month against S3 Standard.

12. Savings left on purpose

Some cheaper options exist and the design does not take them. Each one is listed with its effect, so the Federation can weigh it.

Option	Saving, USD	Why it is not taken
Remove the private endpoints and send center traffic over the internet	43.80 a month	The center traffic would rely on TLS and multi-factor sign-in alone. The endpoints keep it on the private link
CloudFront flat-rate Free plan in place of the pay-as-you-go web firewall	8.00 a month	The plan allows 1 million requests a month, and the public lookup makes about 1.6 million in a harvest month. Above the allowance AWS may serve the traffic from fewer locations, which would slow the lookup in the busiest weeks

Option	Saving, USD	Why it is not taken
Bedrock batch inference at half the token price	About 3.20 a harvest month	Batch jobs may take hours, and a focus confirmation must reach the reviewer the same day
Global inference profile, about 10 percent cheaper on the model	About 0.64 a harvest month	It may process crop images outside the United States, beyond the declared Region
Count the free tiers of CloudWatch, Lambda and Rekognition	A few dollars a month	Free tiers depend on the account and its age, so the estimate assumes none and stays a ceiling

13. Exchange rate and tax

Converting the following year at other rates shows the size of the budget in pesos.

Rate, COP per USD	Following year with the links, COP million
TRM of the price basis, 3,269.88	468.1 to 471.7
Lowest of the last 120 rates, 3,048.12	436.4 to 439.7
Highest of the last 120 rates, 3,796.87	543.6 to 547.8

This conversion keeps every figure fixed in dollars. The lines priced in pesos, the center links, the drone labor and the farm visits, would also change in dollars with the rate, in the opposite direction, so the real effect on a peso budget is smaller than the table shows for those lines.

Tax treatment	Following year with the links, USD
Before IVA	143,164 to 144,268
IVA of 19 percent on every line	170,366 to 171,679

Whether AWS, the operators and the drone contractor charge IVA to the Federation is not confirmed, and import duties on the field hardware and the drones are unknown. The second row is therefore a ceiling for IVA, with duties still outside.

14. Assumptions and pending confirmations

Assumption	Who confirms	What changes if it fails
The TRM of the price basis and figures before IVA	Federation	Section 13 shows the range
A harvest lasts four months and the estimation flights spread over them	Departmental committees	A harvest of another length moves the drone and inference lines in proportion. All flights in one month raise inference to USD 45.95 in that month

Assumption	Who confirms	What changes if it fails
Public sack lookups run at their peak all day in harvest	Federation	The lookup line falls, by less than one dollar
The operator bills by mission hour and uses its staff elsewhere between campaigns	Federation procurement and the operator	A following year rises to about USD 200,738
Operator, insurance, permits and logistics at team estimates	Operator quote and legal review	The drone line moves between USD 75,601 and 188,830 a year
The agrochemical product stays outside the drone cost	Agronomy and the supplier	Fumigation adds treated area times dose times product price
The field items without a price are bought at a reasonable price	Quotes for the enclosure, controller board, probes, mast and cables	Each raises the investment and the reserve
A data plan with coverage and LTE-M service in the three departments	Operator quote	The data and SIM lines of the farms change
The dedicated links cost what the operator reference says	Operator quotes in each municipality	The links move, from USD 64 for shared fiber to 345 at the strongest peso
The team keeps 28 alarms and one dashboard	Platform team	Each alarm moves USD 0.10 a month and each dashboard USD 3.00
Hourly rates of the infrastructure code	Team	The one-time line scales with the rate

15. Sources

- Price basis, for prices, exchange rate and tax.
- Monthly cloud cost, for each cloud line, the environments and the links.
- Inference cost, for the cost per trigger and per 1,000 images.
- Farm hardware and monthly amount, for the investment, the field operation and the proposed amount.
- Drone campaign cost, for the drone service by mission, month and year.
- Observability, operational alarms and dashboards, for the monitoring cost.
- Alert delivery and persistence and archive, for the SMS and storage savings.
- AWS, CloudFront flat-rate pricing plans.
- The figures of the summary are reproduced with `python3 docs/governance/tools/pilot_cost.py --check`, which reads them from the documents above and fails when this document differs.